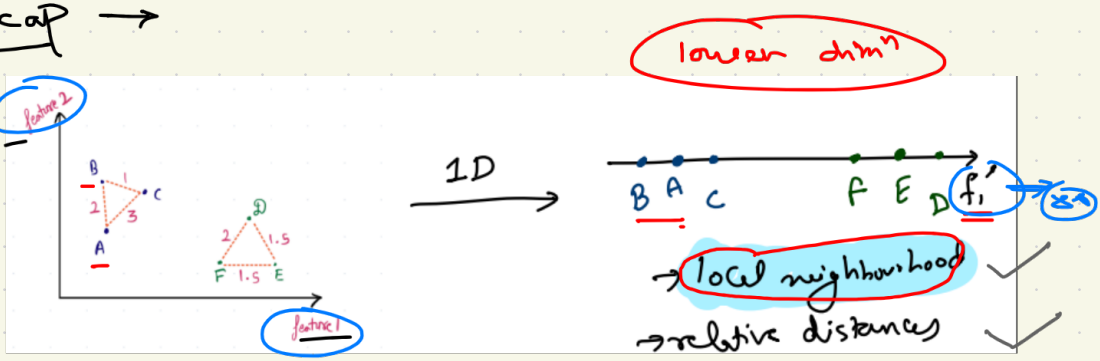
TSNE + Umap ⇒

Agenda → t-SNE] → use case
+

→ Maths
→ Umap

Recap →

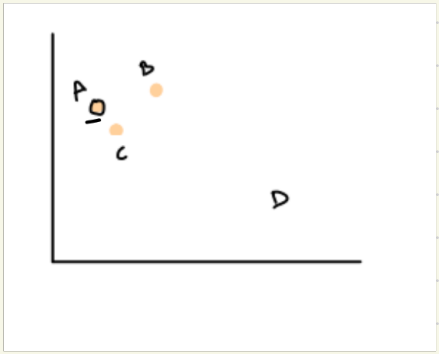


2 dimn ⟶ 1 dimn

$Prob_N(i,j) \approx Prob_N(i,j)$

Higher dimr

Lower dimn



(C) → much closer

=

i & j

$$p_{ij} \propto \frac{1}{dist(i,j)}$$

dist ↑ → low prob

dist ↓ → high prob

$$p_{ij} = \frac{1}{\|x_i - x_j\|} = \frac{1}{d_{ij}}$$

$y = \frac{1}{x}$

$y = \frac{1}{0.0001}$



d_{ij}

dist = v. small

prob → ∞

≥ 1

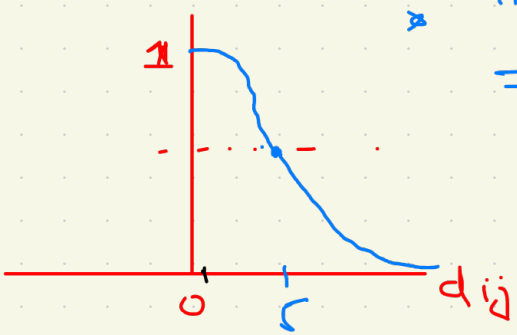
1. $p \to \infty$

2. Prob falling sharply

alternative func^n (ex) d_{ij} Normal dist^n

$$p_{ij} = e^{\frac{-\|x_i - x_j\|^2}{2\sigma^2}}$$

σ × = Spreadness

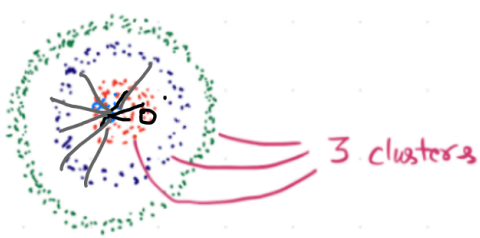


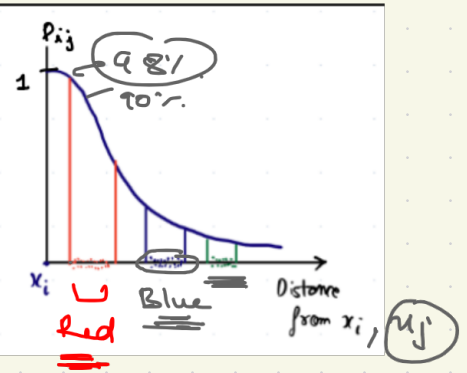
(ex) dist = 0.001 → large prob.

$x_i - x_i = 0$ (1) (ex)

$d \uparrow$

x_i → 0 ×





Recursion

any DP

No Idea

n DP

x_i ⇒ 100 ≤

$p_{i1} + p_{i2} + p_{i3} + p_{i4} + \dots p_{in} = 1?$

independent

$0.98 + 0.90$

No ✗

✗ → 1

2 5 9 1 3

Total = 20

$\frac{2}{20}$ $\frac{5}{20}$ $\frac{9}{20}$ $\frac{1}{20}$ $\frac{3}{20}$

$= 0.1$ 0.25 0.45 0.05 0.15 ⇒

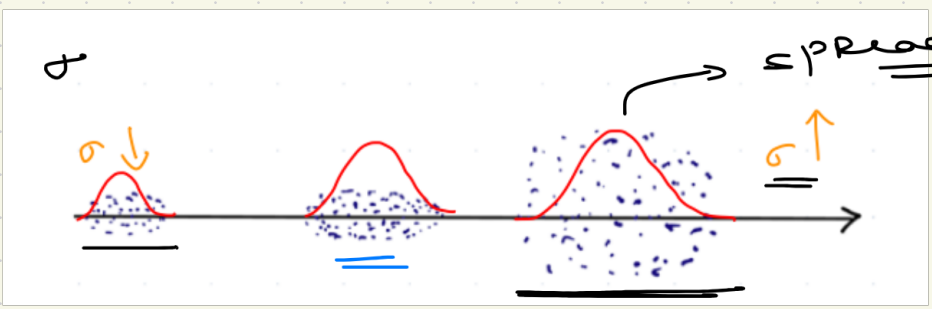
④ Normalize the prob.

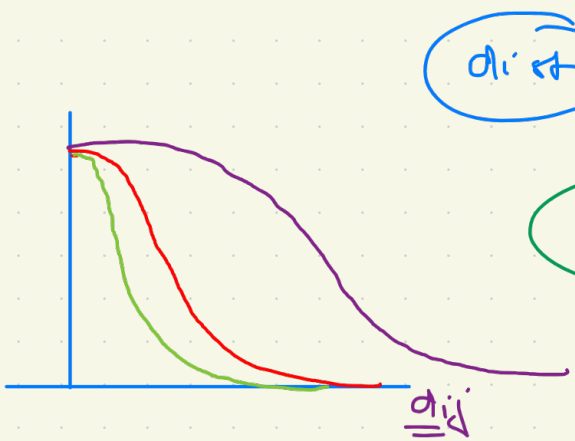
$$p_{ij} = \frac{e^{\frac{-\|x_i - x_j\|^2}{2\sigma_i^2}}}{\sum_{k \neq i} e^{\frac{-\|x_i - x_k\|^2}{2\sigma_i^2}}}$$

→ x_i & x_j

higher dimⁿ

* Normalised prob ⇒





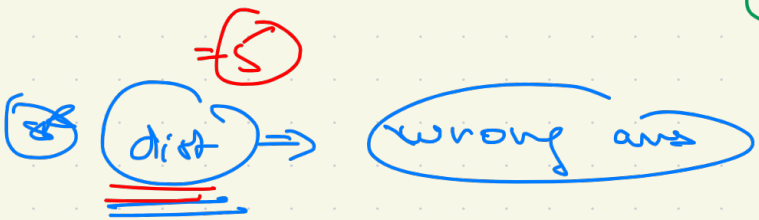


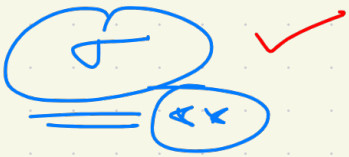
(a) Case 1

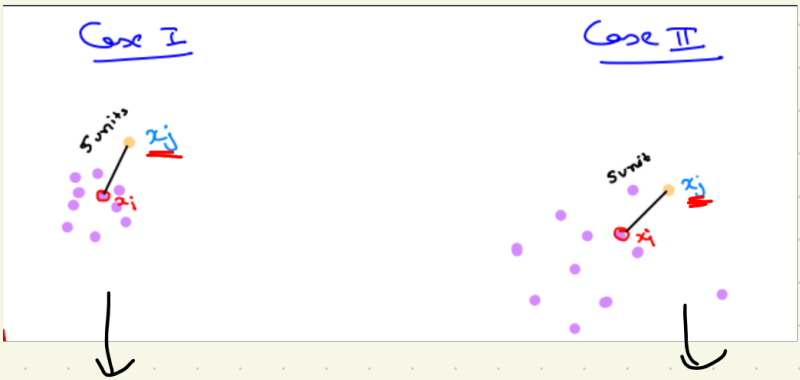
(b) Case 2

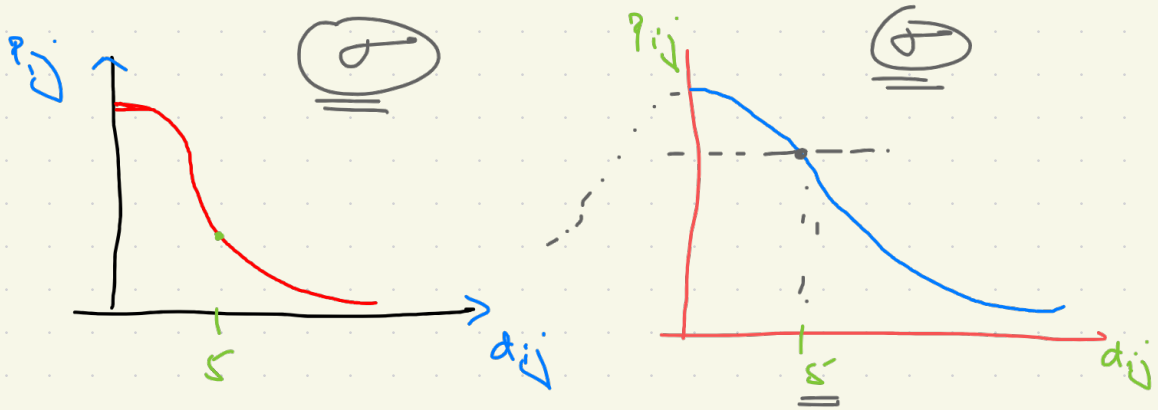
(c) Both

(d) None









dist. → σ ↑

σ ↓

σ ↑

σ very imp → Besides σ ?

Code

data

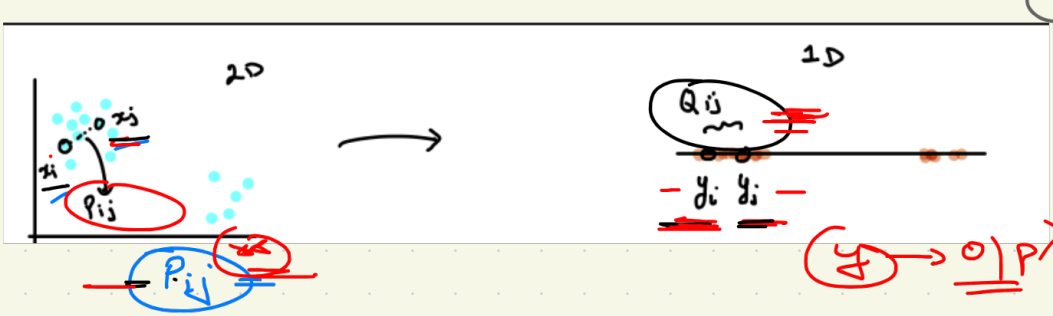
σ → SDP → avg dist

SDP ↓

approx σ

kNN

E



σ → opt ✗

2 dim^n $x_i \longrightarrow y_i$

$q_{ij} \Rightarrow (y_i, y_j)$

$$q_{ij} = \frac{e^{\frac{-\|x_i - x_j\|^2}{2\sigma_i^2}}}{\sum_{k \neq i} e^{\frac{-\|x_i - x_k\|^2}{2\sigma_i^2}}}$$

$\to x_i$ & x_j

higher dim^n

higher dim^n

$\times$ lower dim^n

$$Q_{ij} = \frac{(1 + \|y_i - y_j\|^2)^{-1}}{\sum_{k \neq i} (1 + \|y_i - y_k\|^2)^{-1}} = \frac{\frac{1}{1 + d_{ij}^2}}{\sum \left(\frac{1}{1 + d_{ij}^2}\right)}$$

Q_{ij} — lower dim^n

t distribution with dof = 1

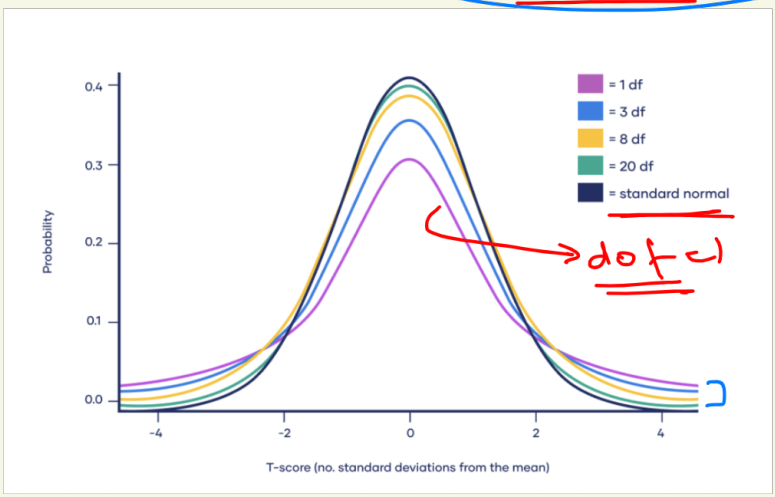
higher → diff

lower → diff

normal → 2 pt. Higher

Diff. b/w Students T dist^n & Normal Dist^b

fatter tails



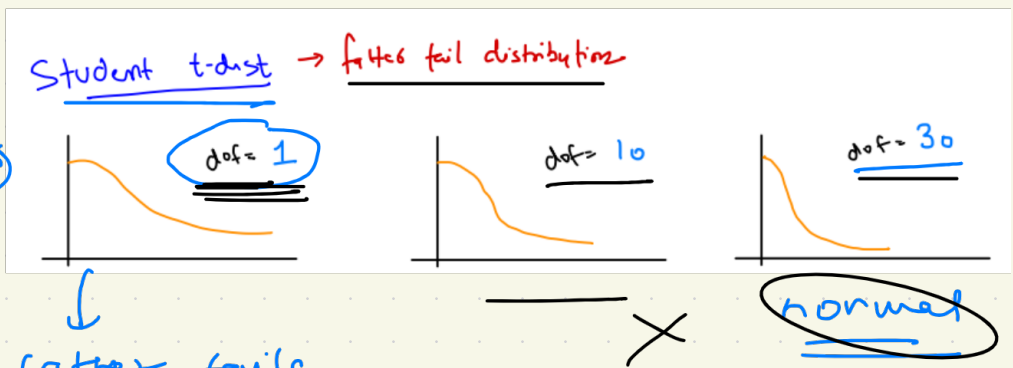
n > 30

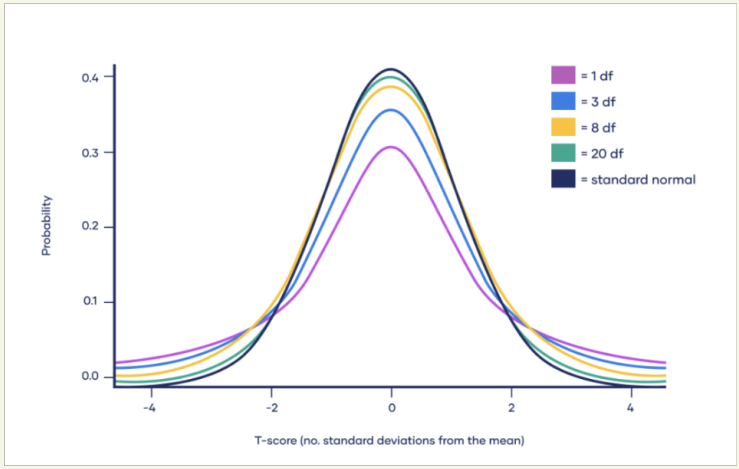
t ≈ Z dist^n

Student t dist^n → fatter tail dist^n

gaussian → μ, σ

t dist^n → 1 param ⇒ dof





data → 3D, 5D → gaussians

2D → T distr with df

CROWDING

CUBE (3D)

SQ

Total space

a = 5

$a^3 = 125$

transform into 2D

keep same size

$a^2 = 25$

less space available

Crowding problem

stretch

Total space = volume = a^3

1mm DP

a = 5

ND

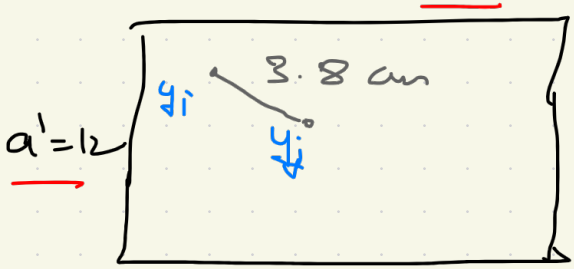
CROWDED

less space

2 cm

Stretched

$a' = 12$



Bigger sheet

Area = 144

more space

$dist(x_i, x_j)$ < $dist(y_i, y_i)$

Higher dim

* in lower dim we have to stretch space to solve for crowding problem

$$dis(x_i, x_j) < dis(y_i, y_j)$$

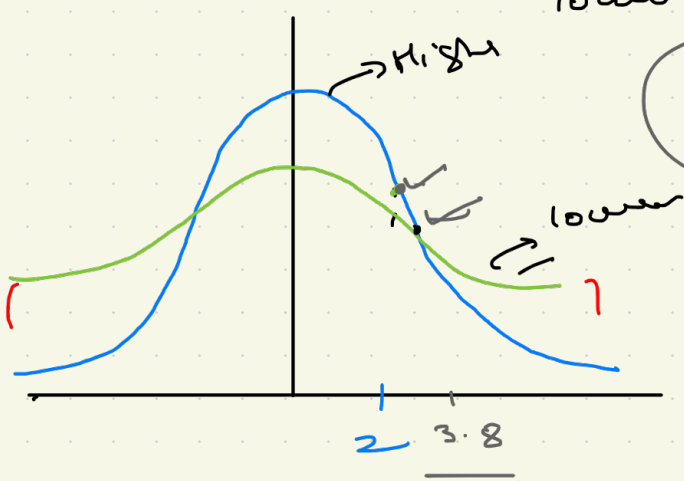
$p_{ij} \approx q_{ij}$

Same dist

single

higher dim^n → $d = 2$ cm

lower dim^n → $d = 3.8$ cm



$p_{ij} \neq q_{ij}$

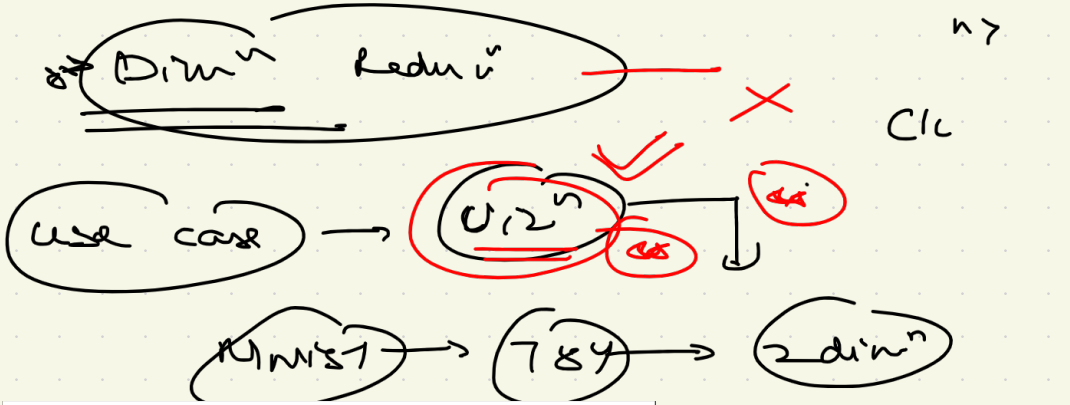
$p_{ij} \approx q_{ij}$

Higher dim^y

gaussian

Lower dim^n

t dist^y



Quiz time!

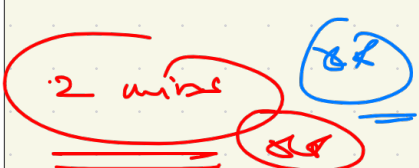
Time Left: 13s

In t-SNE, why is the t-distribution preferred over the normal distribution for lower-dimensional data?

16 users have participated

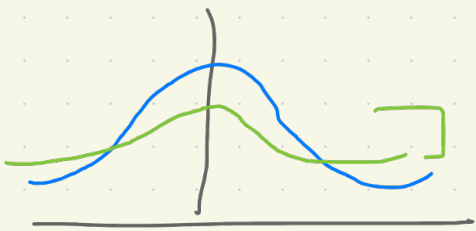
A	The t-distribution has a simpler mathematical form.	6%
B ✓	The t-distribution helps reduce crowding by assigning heavier tails to the data.	75%
C	The t-distribution guarantees a uniform distribution of data points in lower-dimensional space.	19%
D	The normal distribution is not suitable for modeling lower-dimensional data.	0%

End Quiz Now



2 prob. dist^{n}

$p_{ij} \approx q_{ij}$: → similarity?



KL divergence → dissimilarity

New loss

min

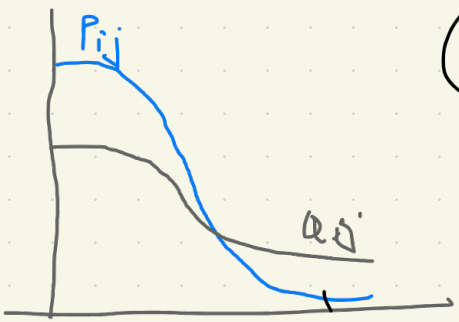
Goal → min. KL divergence

$$KL(P,Q) = \sum_{i=1}^{N}\sum_{j=1}^{N} p_{ij} \log\left(\frac{p_{ij}}{q_{ij}}\right) = 0$$

$\Rightarrow$ loss $\neq 0$

$\rightarrow$ diff everywhere

x_j is far $\rightarrow$



$P_{ij} \approx 0$

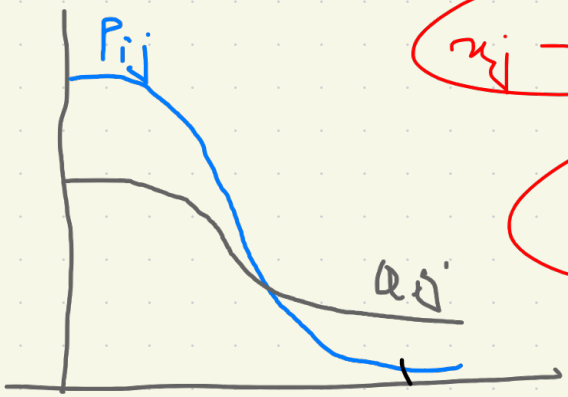
Q_{ij} ✗

nearly 0

$P_{ij} \approx 0$

KL div $= 0$ $\Rightarrow$

x_j closer to x_i



$x_j \rightarrow x_i$

$P_{ij} > Q_{ij}$ ✓

$$\frac{P_{ij}}{Q_{ij}} > 1$$

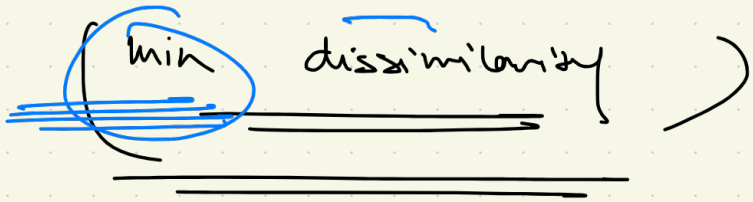
$$\log\left(\frac{P_{ij}}{Q_{ij}}\right) > \log(1) \rightarrow 0$$

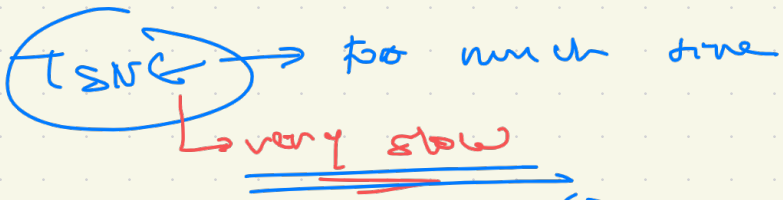
log → x −ve

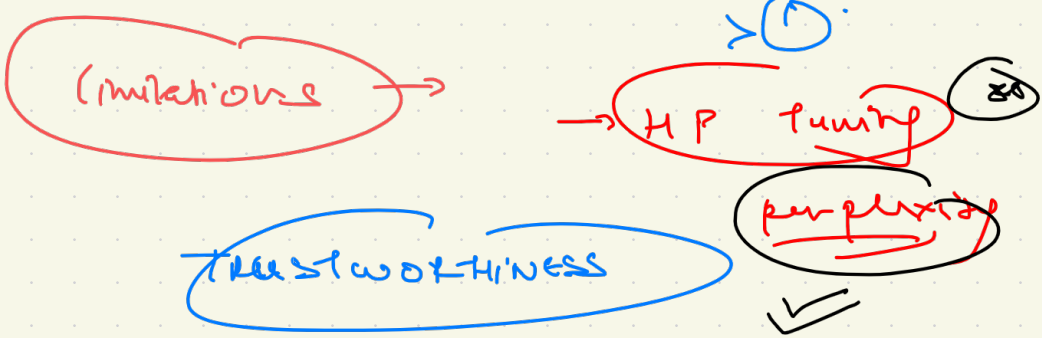
$P_{ij} > Q_{ij}$ for smaller dists

KL div → never becomes −ve

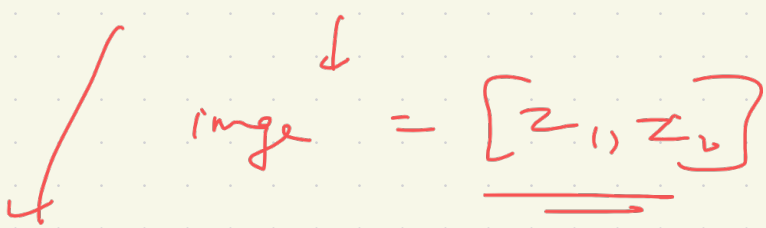
KL divergence ✓


min
dissimilarity


tSNE
too much time
very slow


limitations
HP tuning
perplexity
TRUSTWORTHINESS


img
=
[x1 x2 --- x64]


img
=
[z1, z2]

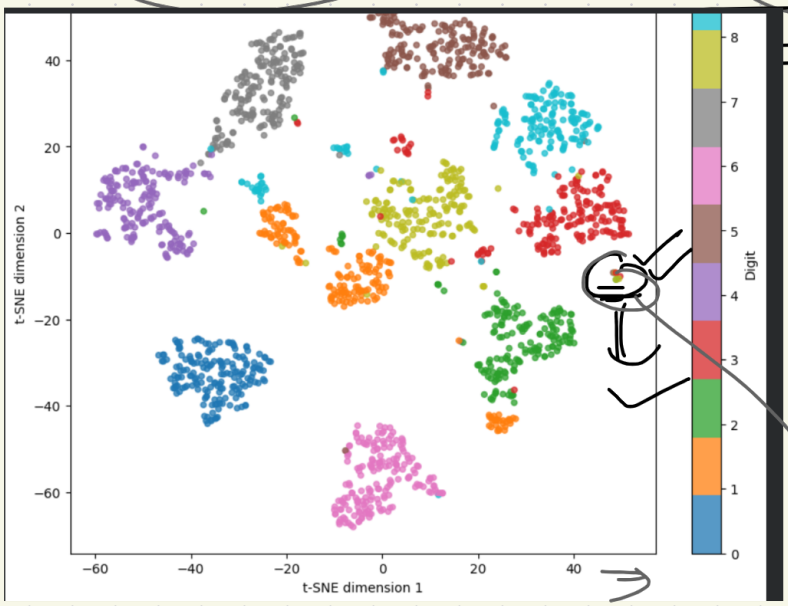

98%
2D
8%
moderate

MSE

R^2

TSNE

Umap



outlier

local

Anomaly detection

tSNE

Umap

Vizu

X → KD